ModelIt!

Teacher's Guide

Why Do Environmental Inequities Exist?

Modeling Pollution Exposure Disparities, Environmental Justice, and Systemic Solutions

Quick Reference

Grade Level	Grade 12L2
NGSS Standard	HS-ESS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-3

Create a computational simulation to illustrate the relationships among the management of natural resources, the sustainability of human populations, and biodiversity.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct evidence-based explanations for environmental inequity patterns using their computational models, and design comprehensive solutions that address systemic drivers rather than individual symptoms.
Disciplinary Core Idea	**ESS3.C Human Impacts on Earth Systems** — The sustainability of human societies depends on the equitable management of natural resources and environmental quality. When environmental burdens are disproportionately concentrated in specific communities, it undermines both social stability and the principles of sustainable development.
Crosscutting Concept	**Systems and System Models** — Students model environmental inequity as a complex system with feedback loops that maintain disparities, understanding that effective solutions must address multiple system components simultaneously rather than treating individual symptoms in isolation.

CAST Testing Alignment

MULTIPLE CHOICE:

A low-income community has 5 times more pollution sources per square mile than the nearby affluent suburb. The state implements equal environmental enforcement across all communities. Based on your model, what is the most likely outcome? A) Pollution exposure becomes equal across communities within 5 years B) Ongoing violations decrease but overall exposure remains higher because existing facilities stay in place C) The affluent community\'s pollution increases to match D) The low-income community gains political power automatically

CONSTRUCTED RESPONSE:

Using your environmental justice model, explain why a community surrounded by a refinery, chemical plants, a highway, and a waste station experiences health impacts far greater than what any single pollution source would cause. Use the concept of cumulative impact and reference at least three model components including the feedback loops that maintain the disparity.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Environmental justice research has consistently demonstrated that environmental hazards are not randomly distributed across communities. Low-income communities and communities of color bear disproportionate burdens of pollution, toxic contamination, and environmental degradation — a pattern that holds across every region, every city, and nearly every country studied. This is not coincidence but the predictable result of systemic forces that can be identified, modeled, and ultimately changed.

The Data: Environmental Inequity by the Numbers

The evidence for environmental inequity is overwhelming and consistent. A landmark 2018 EPA study found that Black Americans are exposed to 54% more particulate matter pollution from industrial sources than the overall population. A 2021 study in Science found that communities of color breathe 56% more pollution from fine particulate matter than they produce through their consumption. Superfund toxic waste sites are disproportionately located in communities of color. Residents within 3 miles of a hazardous waste facility have income levels 15-25% below the national average. In Los Angeles, neighborhoods that were redlined in the 1930s have 50% fewer trees and 3x more industrial facilities today than neighborhoods that were rated favorably. These patterns exist in every US state, most US cities, and across international borders — they are systemic, not anecdotal.

How Environmental Inequity Was Created

Environmental inequity was built through decades of policy decisions. Zoning laws in the early 20th century explicitly designated neighborhoods as industrial or residential along racial lines. The Home Owners\' Loan Corporation redlining maps, created in the 1930s, designated Black and immigrant neighborhoods as hazardous investment risks, directing resources away and industrial development toward these areas. Federal highway construction in the 1950s-1960s deliberately routed interstates through Black neighborhoods, destroying community fabric while concentrating transportation pollution. The siting of waste facilities, power plants, and industrial operations consistently followed the path of least political resistance — which meant communities with the least political power, typically low-income communities of color. These were not individual bad decisions but systemic patterns driven by economic incentives and power imbalances.

Feedback Loops That Maintain Inequity

Environmental inequity is self-reinforcing through multiple feedback loops. Pollution lowers property values, which concentrates low-income residents who cannot afford to move, which reduces the community\'s tax base and political power, which reduces environmental enforcement, which allows more pollution and lower property values. Health impacts from pollution reduce work capacity and increase medical costs, which reduces economic opportunity, which reduces the community\'s ability to organize politically, which reduces its ability to prevent new pollution sources. Schools in polluted areas have higher student absenteeism and lower academic performance due to health effects, which reduces educational attainment, which limits career options, which keeps residents in the same community. Breaking these cycles requires interventions that address multiple loops simultaneously.

Environmental Justice Solutions: From Protest to Policy

The environmental justice movement, which began with community protests against toxic waste siting in Warren County, North Carolina in 1982, has grown into a powerful force for policy change. California\'s CalEnviroScreen tool maps cumulative environmental burdens and directs 25% of cap-and-trade revenue to the most impacted communities. The Biden administration\'s Justice40 initiative directs 40% of federal climate investment benefits to disadvantaged communities. Community benefit agreements require that new industrial development provides direct benefits to affected neighborhoods. Right-to-know laws ensure communities can access data about pollution in their neighborhoods. The most effective approaches combine top-down policy (equal enforcement, investment mandates) with bottom-up community power

(organizing, legal advocacy, political participation) to address both the symptoms and the systemic causes of environmental inequity.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Pollution Source Density, Community Socioeconomic Status, Pollution Exposure Level, Health Outcome Disparity, Regulatory Enforcement Level, Community Political Power, and Green Infrastructure Access as key components of the environmental justice system.

E **Establish Relationships**

Students determine that pollution concentration correlates with socioeconomic status and political power, that health impacts from pollution reduce economic opportunity in a feedback loop, that regulatory enforcement varies by community power, and that comprehensive interventions outperform single-factor approaches.

V **Visualize & Model**

Students build a computational model showing how pollution siting, enforcement, community power, and health outcomes interact through feedback loops to create and maintain environmental inequity across communities.

E **Evaluate Outcomes**

Students run status quo, equal enforcement, and systemic transformation scenarios to test predictions about which interventions most effectively reduce environmental disparities.

R **Revise & Extend**

Students add historical zoning legacy, green gentrification risk, or cumulative impact screening to explore how additional factors affect the dynamics and solutions of environmental inequity.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Your zip code is a better predictor of your health than your genetic code. Today we\'re modeling why — and building solutions."*

Watch For

Show cover image. Ask: What differences do you notice between these two neighborhoods in the same city? What factors might explain those differences? Take responses without correcting — the model will provide evidence.

Slide 2: Learning Objectives

3 min

Say: *"Today we\'re using science to understand a social reality: environmental burdens are not distributed equally. Our model will reveal the systemic mechanisms that create and maintain these patterns."*

Watch For

Have students read objectives. Pre-teach vocabulary. Key concept: cumulative impact — it\'s not just one pollution source, it\'s the total burden from everything combined. This is why some communities face health crises while neighbors do not.

Slide 3: Big Question

3 min

Say: *"If clean air and water are basic human needs, why do some communities breathe dirtier air and drink less safe water than others? The answer is not random — it\'s systemic. Write your hypothesis."*

Watch For

Quick-write: Students write their initial explanation for why environmental burdens are unequally distributed. Save for comparison after modeling reveals the systemic feedback loops.

Slide 4: LEVER Framework

2 min

Say: *"We\'ll model environmental inequity as a system — not a collection of individual bad luck. The model will reveal feedback loops that maintain disparities and leverage points that can break them."*

Watch For

Preview LEVER steps. Emphasize: environmental inequity is not random chance — it\'s a system with identifiable mechanisms. Understanding the system is the first step to changing it.

Slide 5: Activity 1: Components

8 min

Say: *"Which components are external drivers of the environmental justice system, and which are internal responses? Think about what sets up the inequity versus what results from it."*

Watch For

Guide sorting: External = Pollution Source Density (historical siting decisions), Community Socioeconomic Status (economic structure). Internal = the five responses. Discuss: These externals were created by policy decisions — they are not natural or inevitable.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the feedback loops: pollution affects health, health affects economic opportunity, economic status affects political power, political power affects enforcement, enforcement affects pollution. Where does the cycle feed back on itself?"*

Watch For

Students draw arrows. Key discovery: the self-reinforcing cycle of disadvantage. Ask: Once this cycle is running, can it stop on its own? (No — it requires external intervention to break.) Which component, if changed, would disrupt the most feedback loops? (Community Political Power.)

Slide 7: Activity 3: Simulation

10 min

Say: *"Run all three scenarios. Pay attention to why equal enforcement alone isn\'t enough — and what the comprehensive approach achieves that single interventions cannot."*

Watch For

Students run scenarios. Key discussion: Why does equal enforcement reduce violations but not close the exposure gap? Because the physical pollution sources remain concentrated. Ask: What does it take to achieve actual equity, not just equal rules?

Slide 8: Discoveries

5 min

Say: *"Environmental inequity isn\'t random and it isn\'t inevitable. It was created by systems and it can be changed by systems. But changing it requires understanding — and addressing — the feedback loops that maintain it."*

Watch For

Class discussion. Revisit initial hypotheses. Key question: Now that you\'ve seen the systemic dynamics, what surprises you about how environmental inequity works? What\'s the most important insight for creating change?

Slide 9: STEM Challenge

12 min

Say: *"A community of 30,000 people surrounded by pollution sources has secured $50 million for environmental justice. Design the plan. Every dollar must be justified by your model data. Who benefits, and how do you prevent displacement?"*

Watch For

Groups design comprehensive environmental justice plans allocating $50 million across pollution reduction, health services, community empowerment, green infrastructure, and anti-displacement measures. Present with model evidence for every allocation decision.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Pollution Source Density, Community Socioeconomic Status
- **INTERNAL:** Pollution Exposure Level, Health Outcome Disparity, Regulatory Enforcement Level, Community Political Power, Green Infrastructure Access

Relationships

- **Pollution Source Density to Pollution Exposure Level:** POSITIVE (+) — More pollution sources in a community directly increase the concentration of pollutants residents breathe, drink, and contact. Each additional source compounds the cumulative burden.
- **Pollution Exposure Level to Health Outcome Disparity:** POSITIVE (+) — Higher pollution exposure causes higher rates of respiratory disease, cardiovascular problems, cancer, developmental delays, and premature death. Health disparities between communities widen as exposure differences increase.
- **Health Outcome Disparity to Community Socioeconomic Status:** NEGATIVE (-) — Poor health reduces work capacity, increases medical costs, lowers educational attainment through absenteeism, and reduces overall economic opportunity, suppressing the socioeconomic status of the affected community.
- **Community Socioeconomic Status to Community Political Power:** POSITIVE (+) — Higher socioeconomic status provides greater access to political participation, legal resources, media attention, and organizational capacity. Wealthier communities are better able to oppose polluting facilities and demand environmental enforcement.
- **Community Political Power to Regulatory Enforcement Level:** POSITIVE (+) — Communities with greater political power successfully demand more frequent inspections, stricter violation penalties, faster cleanup timelines, and stronger opposition to new pollution sources. Political power translates directly to environmental protection.
- **Regulatory Enforcement Level to Pollution Exposure Level:** NEGATIVE (-) — Stronger enforcement reduces pollution by penalizing violations, requiring emission reductions, and compelling cleanup of contaminated sites. Communities with robust enforcement have measurably lower pollution exposure.

Key Discoveries

- Environmental inequity is maintained by feedback loops — health impacts from pollution reduce economic opportunity, which reduces political power, which reduces regulatory enforcement, which allows more pollution, creating a self-reinforcing cycle that concentrations of disadvantage without external intervention
- Equal enforcement alone is insufficient to reverse environmental inequity because the physical pollution sources remain concentrated in disadvantaged communities — equal enforcement reduces ongoing violations but does not relocate existing facilities or address cumulative historical contamination
- Community Political Power is the strongest leverage point in the model — when communities gain genuine decision-making authority over facility siting, zoning, and enforcement priorities, they can prevent new pollution sources and demand cleanup of existing ones, breaking the feedback loop at its most effective point
- The most effective intervention is comprehensive — combining pollution source reduction, equal enforcement, community empowerment, and green infrastructure investment produces dramatically larger improvements than any single intervention, because it addresses multiple feedback loops simultaneously

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Environmental inequity is just because poor neighborhoods have cheaper land near industrial areas"	While land costs play a role, this explanation reverses the causation. In many cases, pollution sources were sited in neighborhoods that were already disadvantaged by discriminatory housing policies, not the other way around. Studies show that pollution facilities are disproportionately sited in communities of color even after controlling for income and land cost. Furthermore, the cheap-land explanation treats the inequity as natural market forces when it is actually the result of zoning decisions, permitting processes, and political power differentials that systematically favor placing burdens on communities with less power to resist. The model shows that market forces operate within a system structured by policy — change the policy, and the market outcomes change.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Everyone faces the same environmental risks regardless of where they live"	Environmental monitoring data consistently shows dramatic differences in pollution exposure based on neighborhood, with differences of 2-5x in particulate matter, toxic chemical exposure, and contaminated site proximity between communities in the same city. These are not small variations — they translate to measurable differences in asthma rates (2-4x higher), cancer incidence (elevated), and life expectancy (10-20 years shorter in the most polluted communities). The model demonstrates that these differences are not random but are maintained by feedback loops that concentrate pollution in specific communities.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Individual choices like eating well and exercising can overcome pollution exposure"	While personal health behaviors matter, they cannot overcome the physiological effects of breathing polluted air, drinking contaminated water, and living near toxic waste. A child cannot exercise their way out of lead exposure. An athlete who trains outside in polluted air absorbs more pollution, not less. Studies show that zip code — a proxy for environmental quality — is a stronger predictor of health outcomes than individual behavior, diet, or even genetic risk factors. The model demonstrates that environmental conditions set the baseline from which individual choices operate — and in overburdened communities, that baseline is significantly worse.